faisalm93 at DialectSentEval 2026: A Five-Fold MARBERTv2 Ensemble for Arabic Dialect Sentiment Analysis

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Abstract

We describe our submission to DialectSentEval 2026 Subtask 1, a three-class sentiment task covering Moroccan, Egyptian, Lebanese, Jordanian, and Saudi Arabic. Our system fine-tunes MARBERTv2 in five stratified folds and averages their test probabilities. Using raw text without explicit dialect conditioning or external task-specific data, the official submission obtained 0.7939 macro-F1, 0.8152 accuracy, 0.7917 macro precision, and 0.7987 macro recall, ranking eighth among twelve systems in the supplied snapshot. In post-submission MARBERTv2 validation, final-epoch mean fold macro-F1 decreases from 0.9038 with row-stratified folds to 0.8095 when connected components linking repeated ID–label groups and normalized duplicate texts are kept within folds, an absolute difference of 0.0944. These internal validation scores do not alter the official test result. Best-epoch selection, normalization, and dialect tags provide no material gain under the strict protocol.

1 Introduction

Dialectal Arabic sentiment analysis must handle non-standard spelling, code-switching, regional vocabulary, negation, and sentiment-bearing emoji or elongation (Shi and Agrawal, 2025). DialectSentEval 2026 Subtask 1 evaluates sentence-level sentiment prediction across five Arabic dialects (Ezzini et al., 2026a). The task is important because a model trained on one regional variety may misinterpret expressions from another.

Our strategy is a uniform soft-voting ensemble of five MARBERTv2 models. MARBERT is pretrained on Arabic social-media text and is therefore well matched to dialectal language (Abdul-Mageed et al., 2021). We fine-tune one model per cross-validation fold and average their probability vectors. The system achieved 0.7939 official macro-F1 and placed eighth in the supplied twelve-entry leaderboard snapshot. Because macro precision and macro recall differ by only 0.0070, the aggregate scores do not identify a single dominant error type. Our contribution is twofold: we document the official five-fold MARBERTv2 configuration and result, and we provide a post-submission MARBERTv2 comparison showing that internal validation is substantially sensitive to whether linked instances cross folds.

2 Background

2.1 Task and dataset

For an input sentence x, the system predicts $y \in \{\texttt{negative}, \texttt{neutral}, \texttt{positive}\}$. For example, the constructed Egyptian transliteration *el-film da gamil awi* ("this film is very nice") should receive `positive`. This example illustrates the input–output format and is not taken from the evaluation data.

The official task description identifies the resource as Multi-Dialect-Sent (MDS-3), a balanced dataset of 3,000 Moroccan, Egyptian, Lebanese, Jordanian, and Saudi sentences (Ezzini et al., 2026a). The dataset consists of hotel reviews translated and sentiment-annotated by native speakers of the respective dialects. The actual training artifact contains 1,731 rows: 657 negative, 465 neutral, and 609 positive. It contains Darija (433), Egyptian (432), Jordanian (433), and Saudi (433), but no Lebanese training row. The training artifact therefore provides no supervised Lebanese examples. The artifacts used in our submission comprised 1,731 labeled training instances and 525 test instances, totaling 2,256 examples. We report these artifact-level counts without making assumptions about the allocation of the remaining 744 examples, which could not be determined from the supplied artifacts. We participated only in Subtask 1 and used no explicit dialect feature in the official system.

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2.2 Related work

BERT-style encoders support end-to-end classification without a manually constructed sentiment lexicon (Devlin et al., 2019). Arabic models such as MARBERT improve domain alignment by pretraining on large Arabic corpora with substantial dialectal social-media coverage (Abdul-Mageed et al., 2021). ADOR provides a human benchmark of Arabic-dialect hotel reviews (Alharbi et al., 2025b), while the AHaSIS shared task established a related hospitality-domain sentiment benchmark (Alharbi et al., 2025a). Large language models have also been evaluated on sentiment classification across Arabic varieties (Alharbi et al., 2025c). The related AraSentEval shared task jointly studied multi-dialect sentiment classification and Arabic sentiment swapping (Ezzini et al., 2026b). Within this setting, we document the submitted ensemble and complement accuracy-oriented reporting with an audit of whether repeated and duplicate instances can compromise internal validation.

3 System Overview

3.1 Input representation and model

We initialize `UBC-NLP/MARBERTv2` with a randomly initialized three-label sequence-classification head. Each sentence is converted to a string and passed directly to the model tokenizer. Inputs are truncated or padded to 128 tokens, a compute-conscious setting for the short-review domain; the submitted run did not log the proportion of truncated examples. We apply no Arabic normalization, stemming, or segmentation. Alef and ya variants, ta marbuta, diacritics, tatweel, punctuation, emoji, and elongation are left unchanged. We use no external task-specific corpus, lexicon, or pseudo-labeled data; the only external resource is the pretrained MARBERTv2 checkpoint.

3.2 Five-fold ensemble

We construct five sentiment-stratified folds using seed 42. Four folds are used for training and one for validation in each run. A fresh MARBERTv2 model is trained for four epochs. Let $p_m(y \mid x)$ be the probability assigned by fold model m. The submission uses uniform soft voting:

$$\bar{p}(y \mid x) = \frac{1}{5} \sum_{m=1}^{5} p_m(y \mid x), \qquad \hat{y} = \arg\max_{y} \bar{p}(y \mid x). \tag{1}$$

Setting	Value
Encoder	MARBERTv2
Folds / epochs	5 / 4
Batch size / max length	16 / 128
Learning rate	2×10^{-5}
Weight decay / warm-up	0.01 / 0.10
Random seed	42

Table 1: Submitted system configuration.

This is the only configuration included in the official submission.

4 Experimental Setup

The submitted notebook divides rows through five-fold sentiment-stratified cross-validation; the 525-instance test set is used only to produce the official submission. No separate organizer development set is used. Training uses AdamW (Loshchilov and Hutter, 2019), linear decay with 10% warm-up, and gradient clipping at 1.0. Table 1 lists the main settings.

The implementation uses PyTorch, Hugging Face Transformers, pandas, NumPy, and scikit-learn. Training comprised five MARBERTv2 fine-tuning runs, one per fold. It ran on a Google Colab CUDA device, but the exact software versions, GPU model, and wall-clock runtime were not preserved. Validation accuracy and macro-F1 are computed each epoch, but the archived system uses the final rather than the best validation epoch. The fold-level metric logs and OOF probabilities generated during execution were not retained in the archived submission.

Post-submission inspection revealed only 376 unique IDs. All 376 occur in multiple dialects, 397 of the 596 ID–label groups span multiple dialects, and 204 IDs occur with more than one sentiment label. Because grouping by ID alone would merge label-distinct items, we link rows that share either a `(ID, Sentiment)` pair or normalized duplicate text. Transitive closure produces 591 connected components, the largest containing 12 rows, with no component containing conflicting labels. For the MARBERTv2 follow-up, shuffled `StratifiedGroupKFold` with seed 42 assigns these components to folds of 339, 338, 355, 353, and 346 rows. Leakage checks find no component, ID–label group, or normalized text shared across a training–validation boundary.

The official ranking metric is macro-F1, $F_{1,\text{macro}} = \frac{1}{3}\sum_c F_{1,c}$, which weights the three

Official measure	Score
Macro-F1	0.7939
Accuracy	0.8152
Macro precision	0.7917
Macro recall	0.7987
Rank in supplied snapshot	8 / 12

Table 2: Official test-set performance.

Validation condition	Macro-F1
Row split, raw, final	$.9038 \pm .0203$
Row split, raw, best	$.9040 \pm .0200$
Strict split, raw, final	$.8095 \pm .0170$
Strict split, raw, best	$.8100 \pm .0171$
Strict split, normalized, best	$.8066 \pm .0393$
Strict split, dialect tag, best	$.8078 \pm .0249$

Table 3: Post-submission MARBERTv2 mean fold validation macro-F1 ($\pm$ sample standard deviation).

classes equally. The server also reports accuracy, macro precision, and macro recall.

5 Results and Analysis

5.1 Official submission results

Table 2 reports official test results for submission ID 869067. No post-submission result is mixed into this table.

The seventh-place score is only 0.0014 higher, and the fifth-place score is 0.0072 higher. The leader reaches 0.8667, a gap of 0.0728. Macro recall exceeds macro precision by 0.0070, while accuracy exceeds macro-F1 by 0.0213. This profile confirms that accuracy alone is not a suitable selection criterion, but the available official aggregates do not reveal which classes or error directions account for the difference.

Only the ensemble was submitted to the official server. Because neither a separate single-model test submission nor the fold-level test probabilities were retained, these results document the achieved configuration but do not isolate the gain from probability averaging. Similarly, the higher-ranked systems were not reproduced, so the rank and score gaps above are descriptive rather than controlled comparisons. Gold test labels and submitted-system OOF predictions were unavailable for a MARBERTv2 confusion matrix or dialect-level error analysis.

5.2 Post-submission MARBERTv2 validation

We retrain MARBERTv2 with the submitted hyperparameters to compare the original row-stratified validation design against the connected-component split. Table 3 reports the mean and sample standard deviation of the five fold-level macro-F1 scores. These are post-submission internal validation results, not official test scores.

Using the final epoch, which matches the official checkpoint policy, the mean score decreases from 0.9038 under row stratification to 0.8095 under strict grouping, an absolute difference of 0.0944. The best-checkpoint gain is only 0.0002 for the row split and 0.0005 for the strict split, so checkpoint selection does not explain the split effect. Under strict evaluation, normalization and dialect tagging are slightly below raw text. Because the validation partitions differ, these results demonstrate split sensitivity but should not be interpreted as a numerical correction to the official test score. They are means of fold-level macro-F1 rather than pooled OOF scores.

6 Conclusion

We presented a five-fold MARBERTv2 ensemble for DialectSentEval 2026 Subtask 1. The system uses minimal preprocessing and no explicit dialect conditioning, obtaining 0.7939 official macro-F1 and ranking eighth in the supplied snapshot. Its principal limitations are the lack of preserved system-specific OOF predictions, final-epoch checkpointing, and the susceptibility of the original row-level design to linked instances crossing folds. A post-submission MARBERTv2 comparison confirms this sensitivity: final-epoch mean fold macro-F1 decreases by 0.0944 under connected-component validation. Best-epoch selection, normalization, and dialect tags do not improve the strict-split mean. Future work should isolate the contribution of ensembling on a common labeled test set and preserve OOF predictions for class- and dialect-level analysis.

Limitations

The submitted notebook does not preserve package versions, exact GPU, wall-clock time, fold-level metric logs, MARBERT out-of-fold predictions, or instance-level system errors. The training procedure generated validation probabilities in memory, but they were not retained in the archived artifacts. It also cannot be rerun unchanged because a packaging cell overwrites an input path with generated output. This does not affect the recorded official score, but a release should separate immutable inputs from generated files and add schema valida-

tion. The run also did not record the number of inputs truncated at 128 tokens.

No official single-model test result was preserved, so the contribution of five-fold probability averaging cannot be isolated. The post-submission MARBERTv2 comparison uses one split seed and reports fold-level means from a single training run per fold; it does not provide uncertainty across training seeds. Its OOF predictions were not exported, preventing pooled OOF, class-level, dialect-level, or instance-level analysis. The unavailable gold test labels also prevent a controlled single-model versus ensemble test comparison. Aggregate scores can conceal uneven behavior across dialects and should not support high-impact decisions without human review.

Ethical Considerations

Sentiment predictions may reproduce dialect and class imbalances in the training data. The system should not be used for automated moderation, profiling, or other high-impact decisions without dialect-aware evaluation and human oversight. We include no user identifiers. Any reuse or redistribution of the task data should follow the organizers' licensing and access conditions.

Acknowledgments

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A Submitted-System Reproducibility Details

The archived submission notebook maps `negative`, `neutral`, and `positive` to label IDs 0, 1, and 2, respectively. It seeds NumPy, PyTorch, and all CUDA devices with 42. The submitted procedure is:

1. Read the training and test CSV files and convert each sentence to a string without text normalization or dialect prefixes.
2. Construct five folds with shuffled `StratifiedKFold`, stratifying by sentiment label and using random state 42.
3. For each fold, initialize a fresh `UBC-NLP/MARBERTv2` sequence classifier with three output labels. Tokenize to 128 positions with truncation and fixed-length padding.
4. Train for four epochs with shuffled training batches of size 16. Use AdamW with learning rate 2×10^{-5} and weight decay 0.01, a linear scheduler whose first 10% of steps are warm-up steps, and a gradient-norm limit of 1.0.
5. After the fourth epoch, compute validation and test softmax probabilities. Use the validation probabilities as in-memory OOF predictions and add one fifth of the test probabilities to the ensemble accumulator. The OOF arrays were not written to a retained artifact, and no best-epoch checkpoint is selected.
6. Apply `argmax` to the averaged test probabilities, map the indices back to the three text labels, and export the `ID` and `Sentiment` columns.

Training batches are shuffled, whereas validation and test batches are not. The notebook records validation accuracy and macro-F1 after every epoch and saves the final state of each fold model. It does not preserve deterministic backend settings, package versions, the GPU model, or elapsed training time.

B Strict Post-Submission MARBERTv2 Split Construction

The strict folds are used only for the post-submission MARBERTv2 comparison. They do not alter the official result. We first construct a normalized duplicate key by applying Unicode NFKC normalization, removing tatweel, diacritics, and punctuation, converting to lowercase, and collapsing whitespace. Rows are then joined into connected components when they share either the same `(ID, Sentiment)` pair or the same normalized text. A normalized duplicate with conflicting sentiment labels causes the procedure to stop for manual resolution rather than being silently split. No such conflicting normalized group was observed in the 1,731-row training artifact.

The construction yields 591 connected components, with a largest component of 12 rows. We pass the component identifiers to shuffled `StratifiedGroupKFold` with five splits and random state 42. The fold sizes are 339, 338, 355, 353, and 346. Leakage checks confirm that no component, linked ID–label group, or normalized duplicate crosses a training–validation boundary.

C Scope of the Post-Submission MARBERTv2 Results

The follow-up uses the same encoder, optimizer, sequence length, batch size, learning rate, epoch count, and base seed as the submitted system, with fold-specific seeds $42 + m$ for fold index m. It ran with Python 3.13.15, PyTorch 2.11.0+cu128, and an NVIDIA Tesla T4. Table 3 reports the mean and sample standard deviation of five fold-level macro-F1 values. Best checkpoints are selected by validation macro-F1 within each fold. The notebook did not export the OOF prediction arrays, and its test-data stage was not completed because the expected test filename was unavailable. Consequently, the results do not include a single-model versus ensemble test comparison or MARBERTv2 error analysis and must not be presented as official leaderboard scores.